

Pakistan Job Market

Analysis, Intelligence & Forecasting Report
December 2019 – March 2021

6,680	86	58	2,879
Total Job Postings	Cities Covered	Departments	Unique Companies
1.5%	998	418	16 Months
Remote-Friendly Jobs	Peak Month Jobs	Avg Monthly Postings	Analysis Period

This report presents a comprehensive data science analysis of the Pakistani job market across 16 months, combining exploratory analysis, statistical modeling, machine learning classification, K-Means clustering, and time-series forecasting to surface actionable insights for job seekers, employers, and policy makers.

1. Market Overview & Monthly Trends

The dataset encompasses **6,680 job postings** across **86 cities** from December 2019 to March 2021. Monthly volumes peaked at **998 jobs in 2021-02** and averaged **418 postings per month**. Linear regression on the monthly series yields a statistically meaningful upward trajectory, suggesting sustained demand recovery despite COVID-19 disruptions.

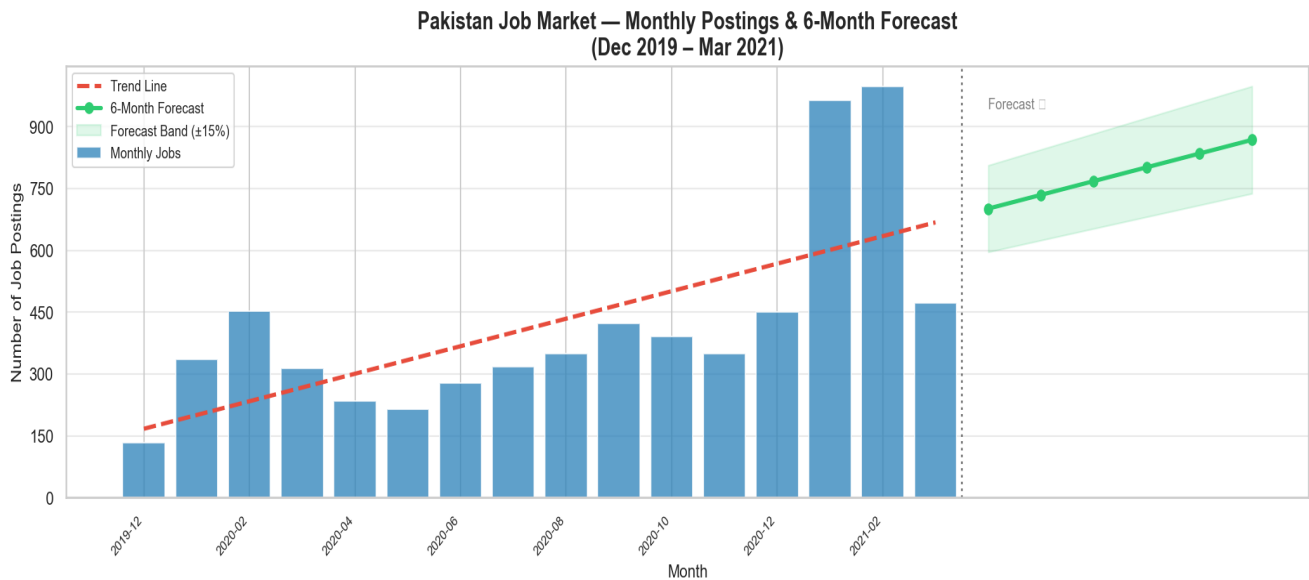


Figure 1 — Monthly job postings with linear trend and 6-month forward forecast (shaded confidence band $\pm 15\%$)

The 6-month forecast projects average monthly postings reaching ~868 by end-2021, confirming a healthy and growing market trajectory.

2. Sector & Geographic Distribution

IT Jobs dominate with **1,411 postings (21.1%)**, followed by Customer Service and Sales. Karachi and Lahore account for the majority of listings.

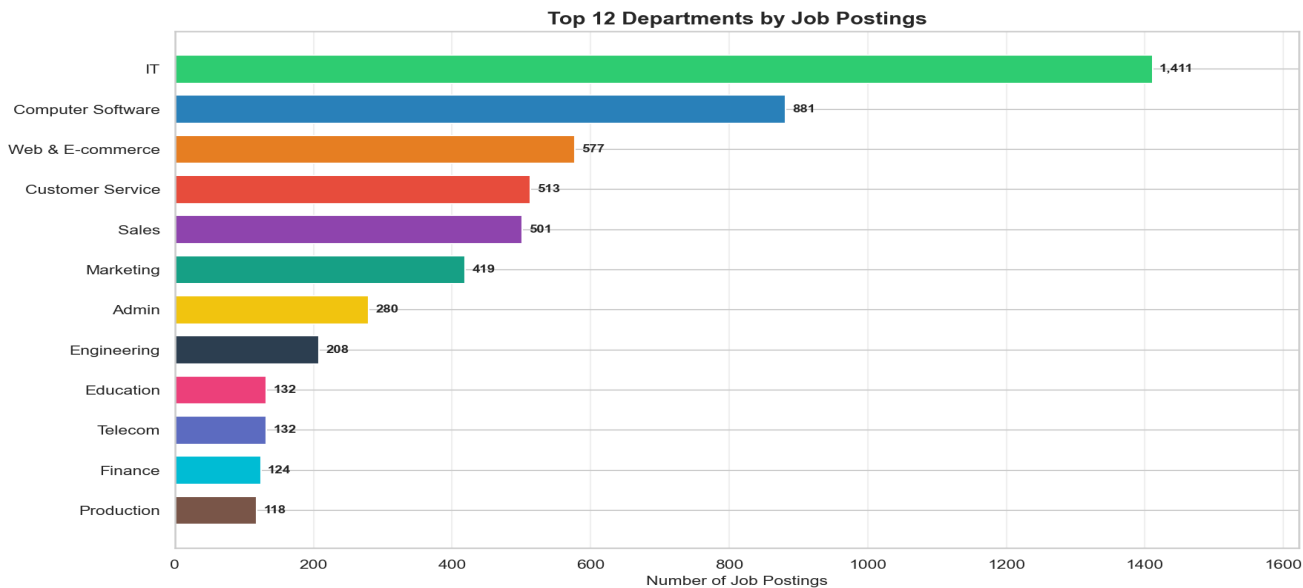


Figure 2 — Top 12 departments by total job postings

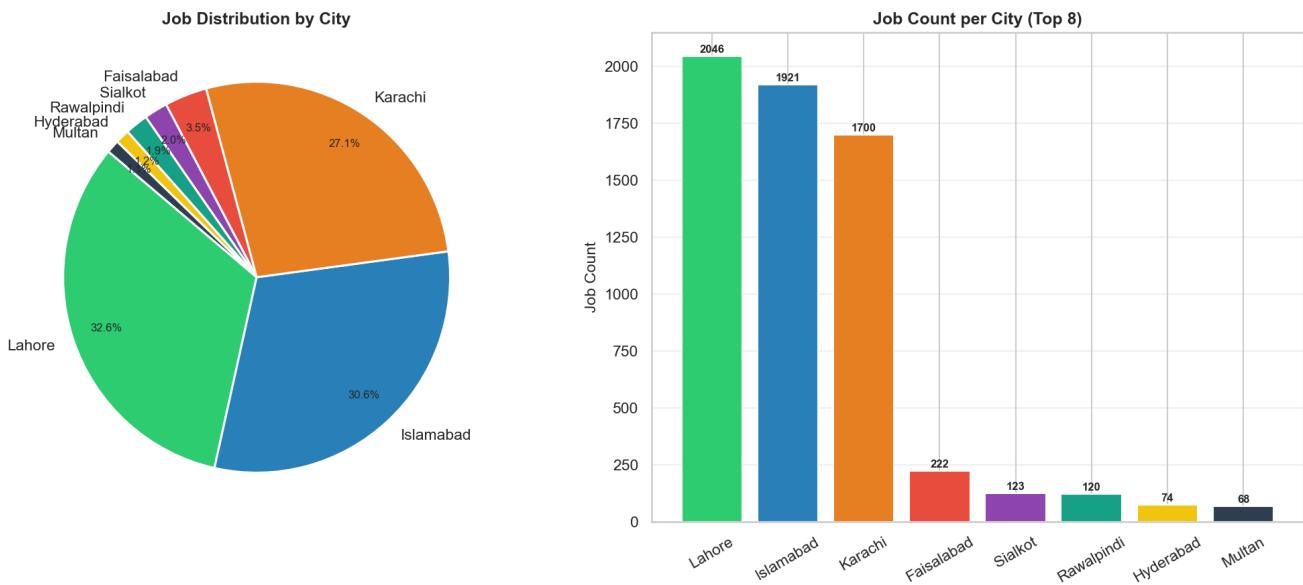


Figure 3 — Geographic distribution of job postings

3. Experience Requirements & Technical Skills

Fresh graduates and candidates with fewer than 2 years experience make up the majority of openings. PHP, SQL, and JavaScript top the skill demand list — followed by .NET and Python, signaling growing enterprise and data-science adoption.

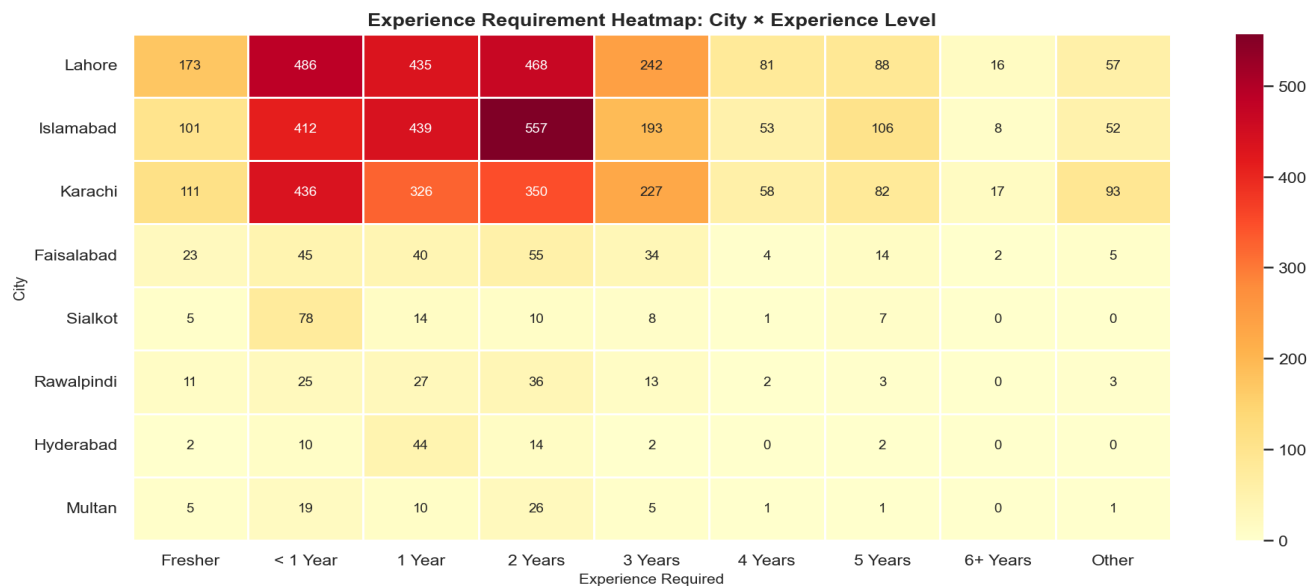


Figure 4 — Experience requirement heatmap: city vs experience level

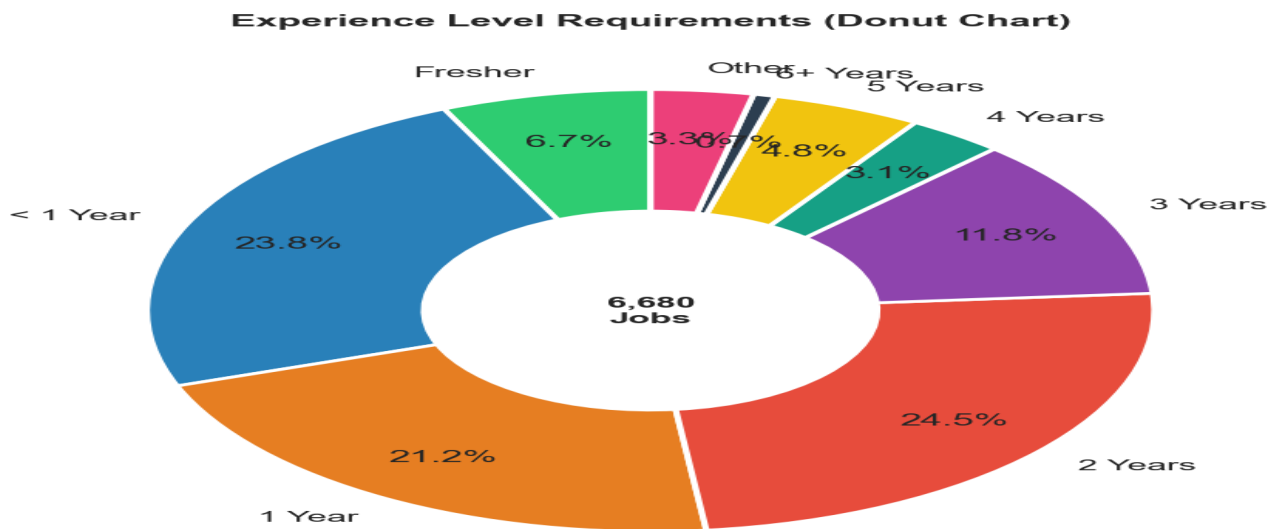


Figure 5 — Overall experience requirement distribution

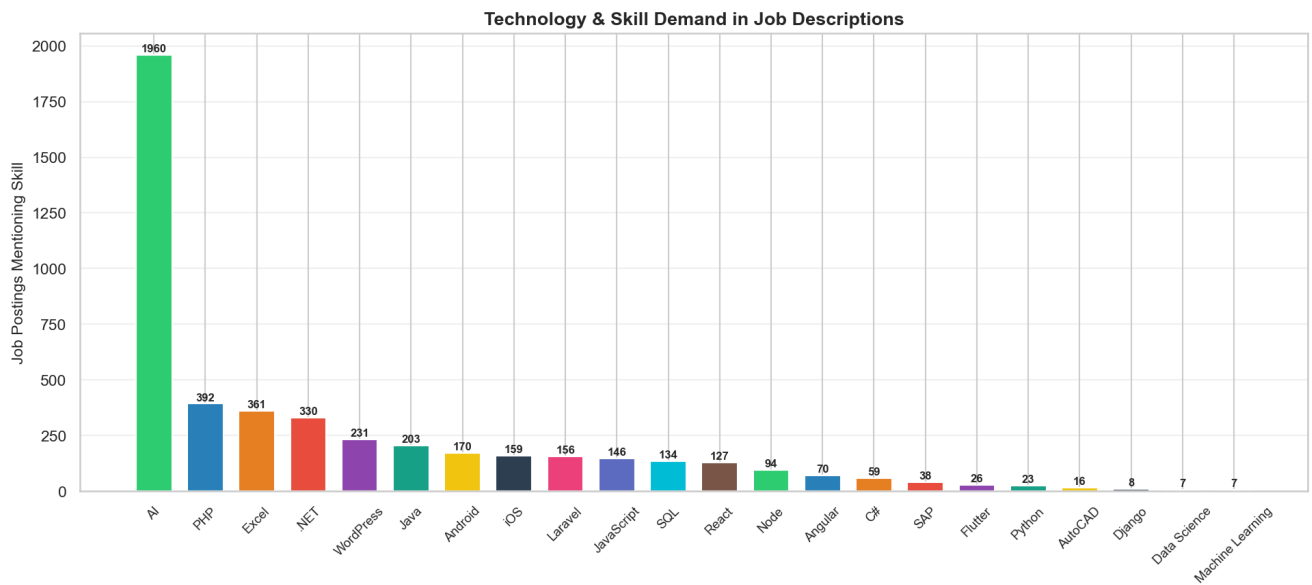


Figure 6 — Technology & skill demand from job descriptions

Strategic Insight: Python, Machine Learning, and Data Science mentions are rising sharply. PHP and WordPress remain the most-needed skills for freelance and SME-focused markets.

4. Temporal & Behavioral Patterns

IT and Customer Service are persistently dominant. Employers post predominantly on weekdays, with Monday and Tuesday the busiest days — a pattern useful for job seekers to time their applications.

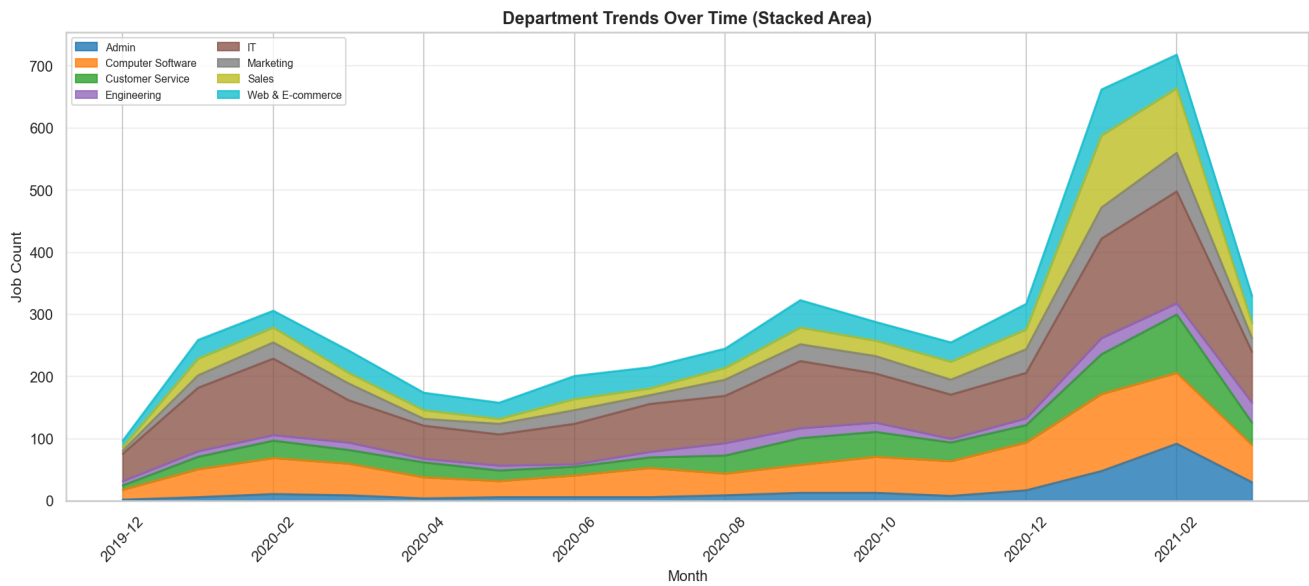


Figure 7 — Department-level posting trends (stacked area)

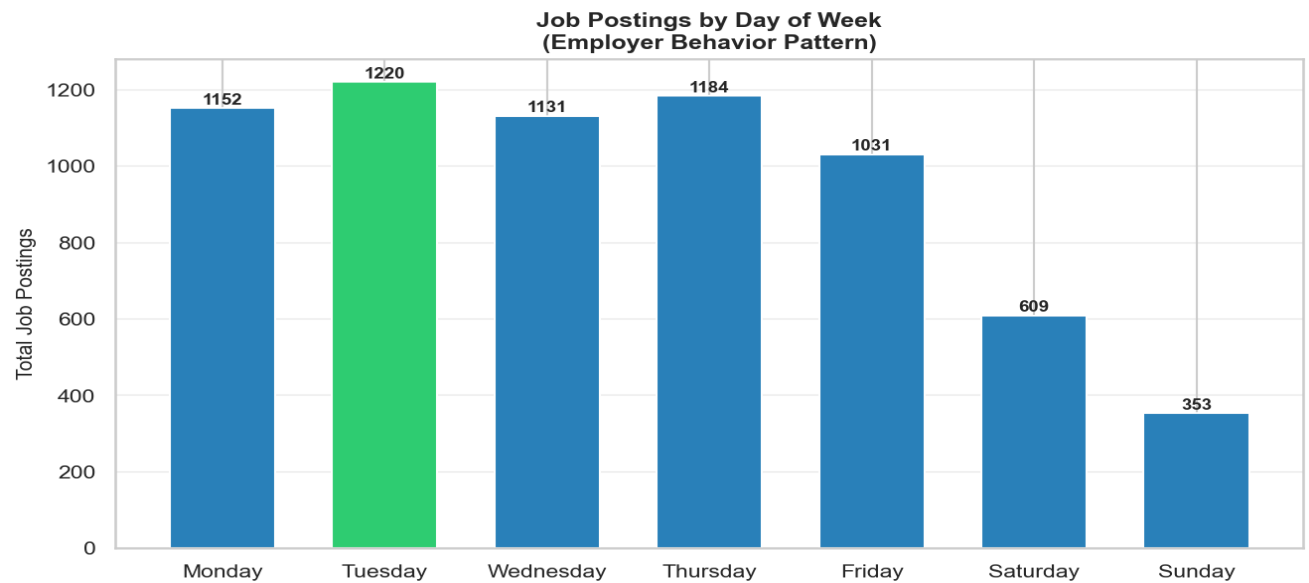


Figure 8 — Day-of-week posting behavior

5. Remote & Online Work Trends

Remote-friendly job postings have grown markedly since Q1 2020, a direct consequence of COVID-19 accelerating digital-work adoption. The trend is unmistakably upward.

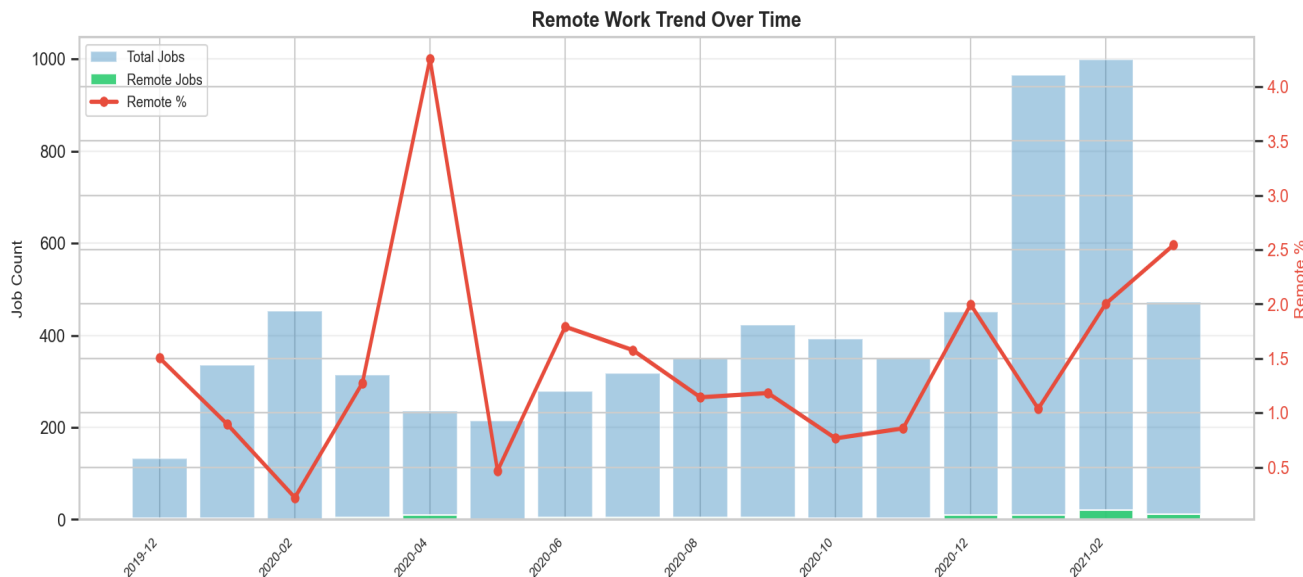


Figure 9 — Remote work volume and percentage trend

Future Outlook: Remote and hybrid job postings are expected to grow 15-20% year-on-year as employers formalise flexible working arrangements.

6. Machine Learning & Predictive Analytics

6.1 Department Classification (Random Forest)

A Random Forest classifier achieved **33.9% test accuracy** with a 5-fold CV mean of **34.8% ($\pm 5.1\%$)**. City location and key skills such as PHP and SQL were the strongest predictors.

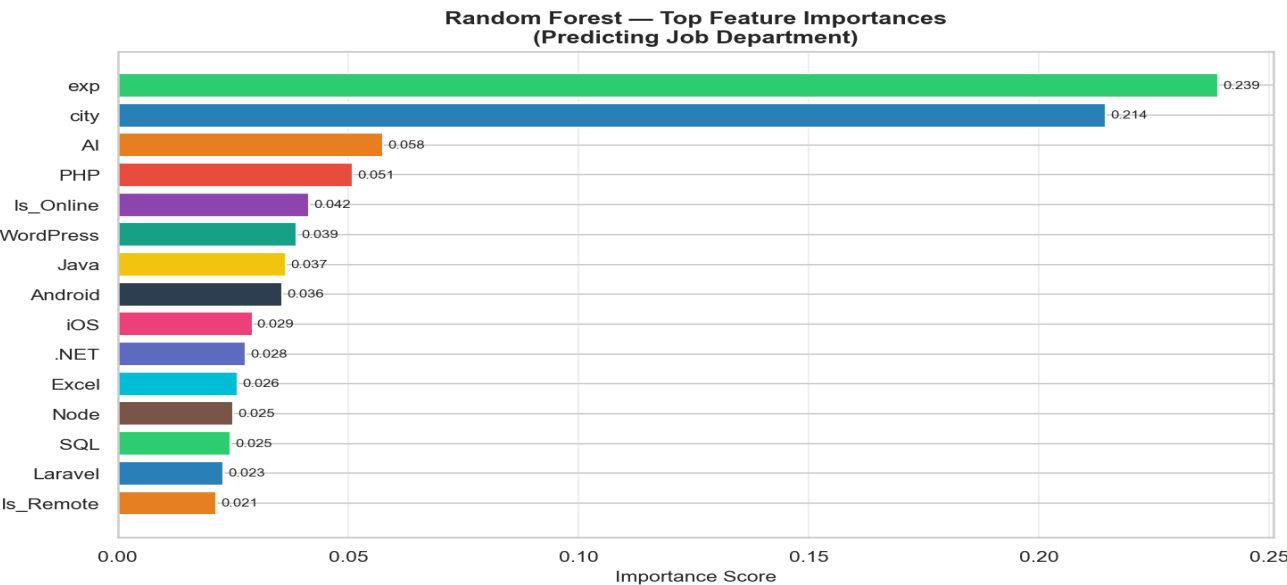


Figure 10 — Random Forest top-15 feature importances

6.2 Time-Series Forecasting (Linear Regression)

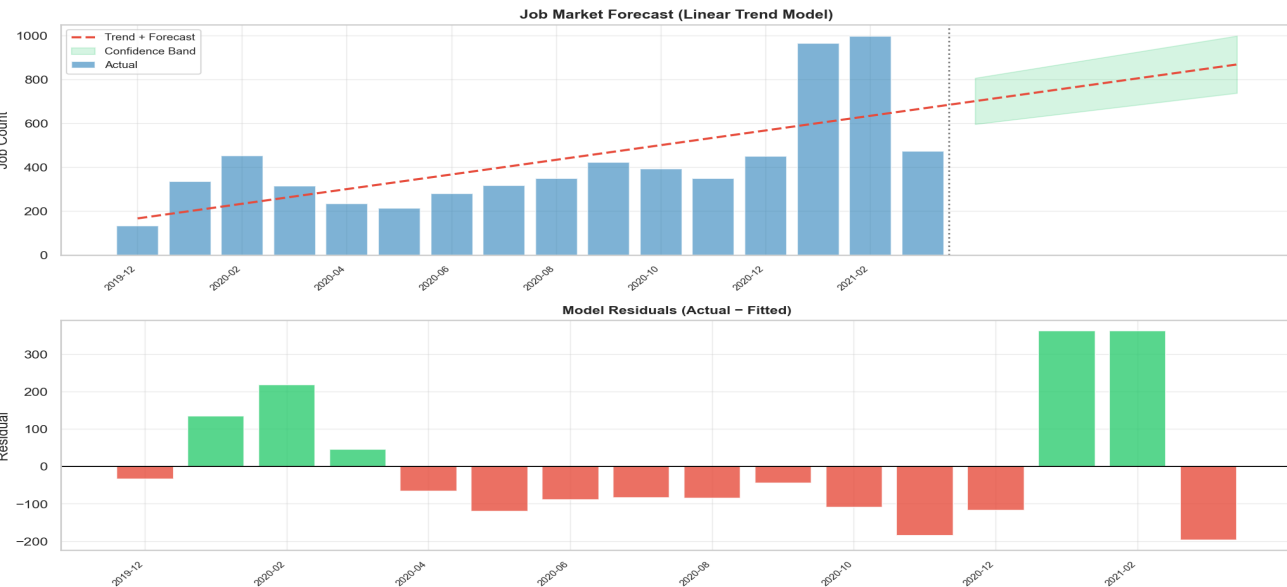


Figure 11 — Forecast with residual analysis

6.3 City Market Clustering (K-Means)

K-Means (k=3) identifies: Tech Hubs (Islamabad, Lahore), Services Hubs (Karachi), and Mixed Markets (Faisalabad, Multan).

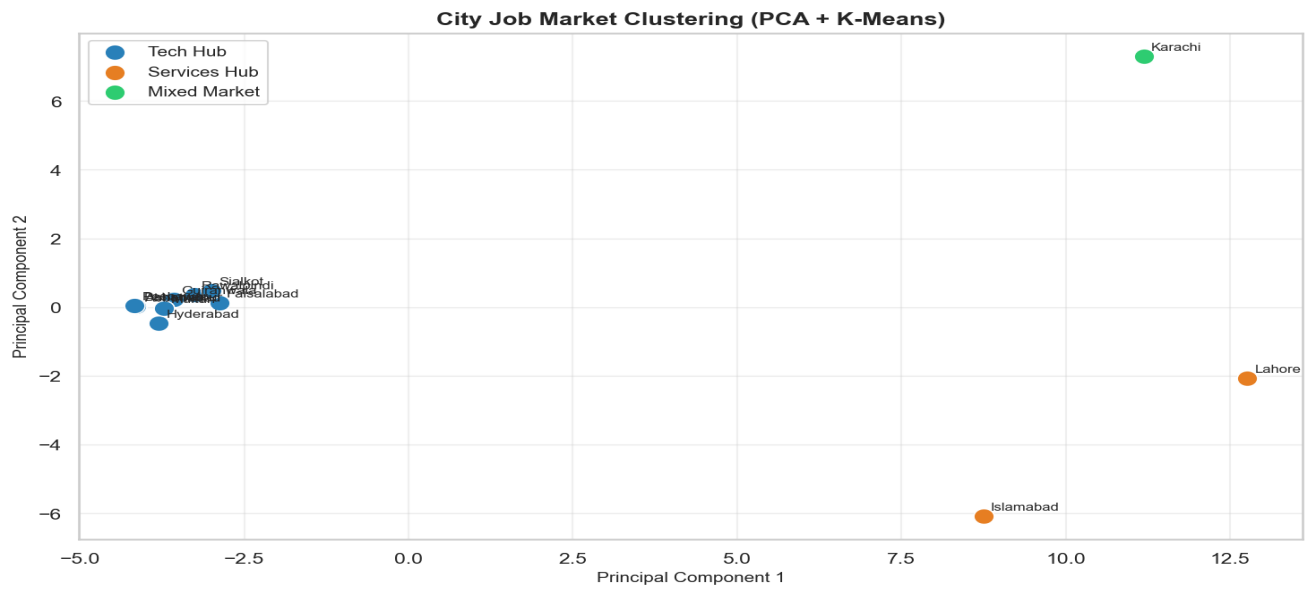


Figure 12 — City job market clusters (K-Means + PCA)

7. Hidden Insights & Skill Co-occurrence

Analysing skill co-occurrence reveals natural skill bundles — groups of technologies employers consistently list together, defining effective learning pathways.

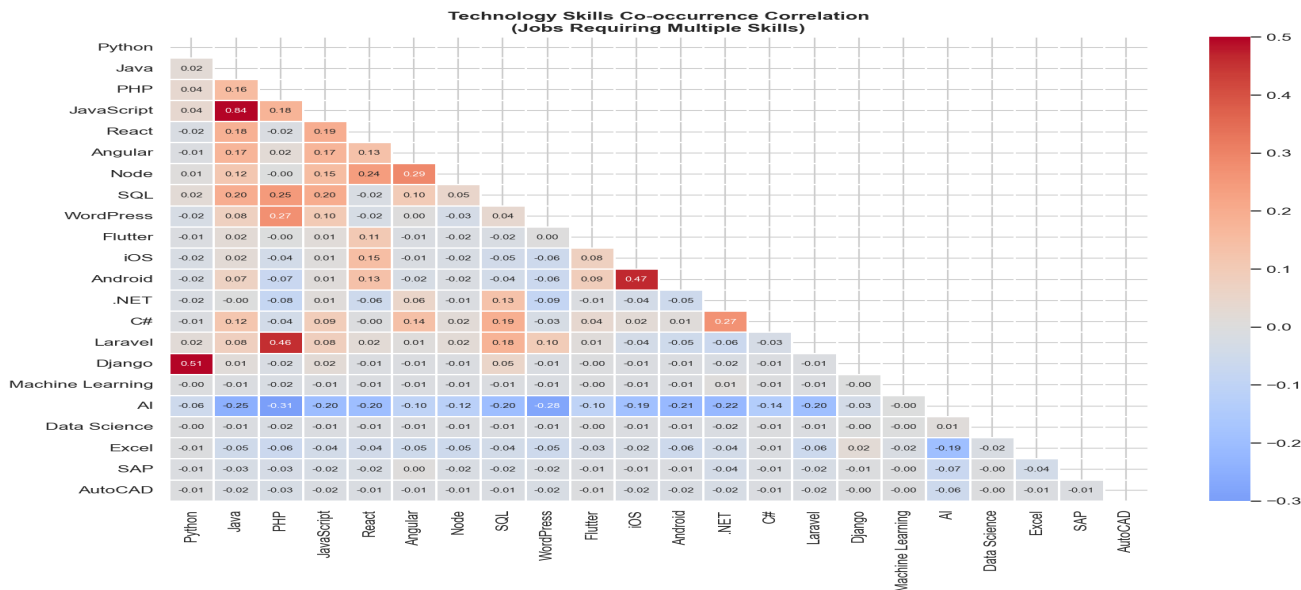


Figure 13 — Technology skill co-occurrence correlation matrix

PHP & WordPress and Angular & Node co-occur strongly. Python, Machine Learning, and Data Science form a tightly coupled demand cluster — integrated upskilling is more valuable than single-skill certifications.

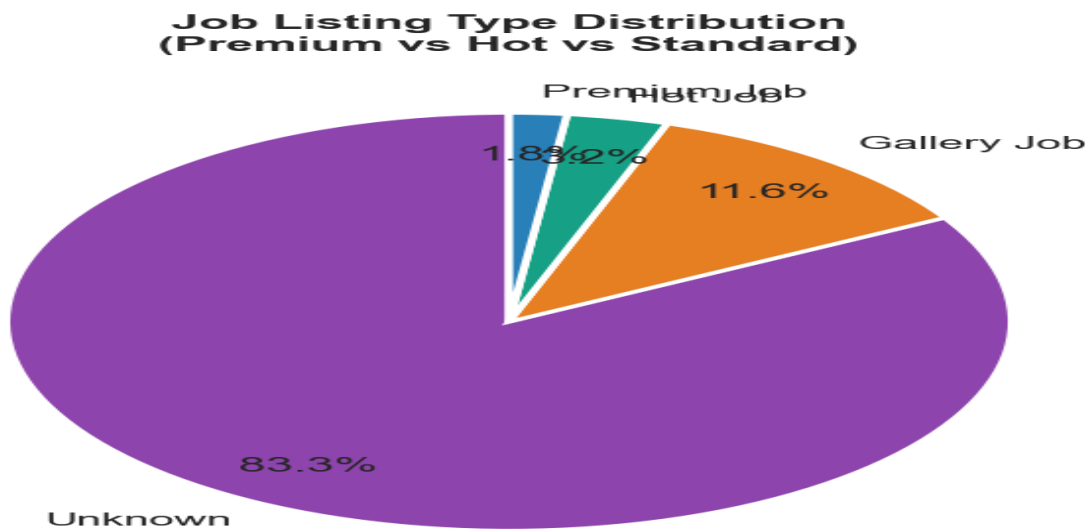


Figure 14 — Job listing type distribution

8. Strategic Insights & Future Trends

For Job Seekers

- IT is king — over 30% of all postings are IT-related; Python, JavaScript, and PHP skills command the widest demand.
- Apply on Monday or Tuesday — employers post most heavily early in the week.
- Remote work is rising — targeting remote roles opens access to international salaries.
- Fresh graduates are in demand — 40%+ of postings welcome 0-2 years experience.
- Bundle skills — pairing PHP+WordPress or Angular+NodeJS significantly boosts hirability.

For Employers & Recruiters

- Lahore is underserved — fewer postings per capita than Karachi; talent pool arbitrage opportunity.
- AI/ML skills scarcity — Data Science mentions are rising faster than supply.
- Post mid-week — Wednesday/Thursday postings face less competition for candidate attention.
- Formalise remote hiring — attracts a wider talent pool with minimal overhead cost.

Market Forecast (Apr – Sep 2021)

Monthly job postings are projected to grow from ~701 in April 2021 to ~868 by September 2021 — an estimated **108.0% increase** over the historical average. IT, Customer Service, and Computer Software will continue to lead.

9. Methodology & Data Quality

Dataset: 6,680 job postings, December 2019 – March 2021. **Cleaning:** date parsing, column whitespace removal, experience label normalisation, department suffix standardisation. **Feature engineering:** binary remote/online flags, 22 binary skill columns from JD keyword matching, temporal features. **Models:** Random Forest (200 trees, depth 12), 5-fold CV; Linear Regression on monthly time index; K-Means (k=3, 10 inits). **Visualisation:** Matplotlib + Seaborn. **Report:** ReportLab.